

SCOUTING INTELLIGENCE BRIEF

The Scouting Engine

Finding a Successor to Mohamed Salah

Task 2.3 • AI Team Training '27 • Phase I

Data: FIFA 19 Complete Player Dataset (Kaggle)

Executive Summary

This brief searches all 18,147 FIFA 19 players to answer one question: **who plays most like Mohamed Salah?** Every player is a 9-dimensional vector (Finishing, ShortPassing, Dribbling, SprintSpeed, Strength, Stamina, Interceptions, StandingTackle, Value). Three metrics — Cosine Similarity, Euclidean Distance, Manhattan Distance — were implemented from scratch and run against Salah’s z-standardized profile.

Standardized, the three metrics agree closely on elite attacking wide players and forwards. Unstandardized, results are dominated by market Value (tens of millions vs. 0–100-scale skills), so “similar” secretly means “similarly priced,” not “similarly skilled.”

Final shortlist: Antoine Griezmann, Leroy Sané, Kylian Mbappé, Cristiano Ronaldo, Coutinho.

1 Methodology

Target: M. Salah (RM) — FIFA 19 profile:

Table 1: Mohamed Salah’s Feature Vector

Fin.	Pass.	Drib.	Speed	Str.	Stam.	Inter.	Tackle	Value
90	82	89	91	70	84	55	43	€69.5M

Two feature matrices were built: raw values, and z-standardized ($z = \frac{x-\mu}{\sigma}$, manual NumPy). All three metrics ran against both, fully vectorized against the whole 18,147-player pool (no per-player loop).

2 Similarity Metrics & Rankings

2.1 1. Cosine Similarity

Angle between vectors, ignoring magnitude — captures profile “shape,” not absolute level.

$$\text{Cosine}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2} \quad (1)$$

2.2 2. Euclidean Distance

Straight-line distance across all 9 dimensions — sensitive to absolute level gaps. Lower = more similar.

$$d_{\text{Euc}}(\mathbf{u}, \mathbf{v}) = \sqrt{\sum_i (u_i - v_i)^2} \quad (2)$$

Table 2: Top 5 — Cosine Similarity (Standardized)

Rank	Player	Position	Score
1	K. Mbappé	RM	0.9968
2	Marco Asensio	RW	0.9953
3	A. Griezmann	CAM	0.9946
4	E. Hazard	LF	0.9939
5	L. Sané	LW	0.9934

Table 3: Top 5 — Euclidean Distance (Standardized)

Rank	Player	Position	Distance
1	Coutinho	LW	1.626
2	A. Griezmann	CAM	1.963
3	J. Rodríguez	LAM	1.973
4	L. Sané	LW	2.015
5	Cristiano Ronaldo	ST	2.075

2.3 3. Manhattan Distance

Sum of absolute differences (L_1) — less swayed by any one large gap than Euclidean. Lower = more similar.

$$d_{\text{Man}}(\mathbf{u}, \mathbf{v}) = \sum_i |u_i - v_i| \quad (3)$$

Table 4: Top 5 — Manhattan Distance (Standardized)

Rank	Player	Position	Distance
1	J. Rodríguez	LAM	3.817
2	A. Griezmann	CAM	3.899
3	K. Mbappé	RM	3.929
4	Coutinho	LW	3.999
5	L. Sané	LW	4.413

2.4 Comparison

Griezmann and Sané appear in **all three** top-5 lists — agreement isn’t a single metric’s quirk. Cosine favors explosive wide attackers of similar “shape”; the distance metrics surface Coutinho and Rodríguez, whose absolute levels sit numerically closest to Salah’s.

3 Reflection: Standardized vs. Unstandardized

Why: Salah’s squared vector magnitude is $\approx 4.83 \times 10^{15}$ from Value alone (€69.5M²) vs. $\approx 47,856$ from all 8 skills combined — Value alone is effectively **100%** of the vector’s length. Without standardization, “nearest neighbor” collapses to “similarly priced,” surfacing goalkeepers and midfielders who share Salah’s transfer value but nothing of his playing style.

4 Final Recommended Shortlist

Ranks averaged across all three standardized metrics (lower avg. rank = stronger match):

Table 5: Effect of Skipping Standardization

Metric (raw)	Top Result	Score
Cosine	Coutinho, Rodríguez, Sané, Silva, Erik-sen	all ≈ 1.0000
Euclidean	Coutinho, then Oblak (GK), Modrić, De Gea (GK)	26.1, jumps to $>1.5M$
Manhattan	Rodríguez, Coutinho, then Oblak, Modrić, De Gea	60–66, jumps to $>1.5M$

Table 6: Consolidated Shortlist

Rank	Player	Position	Avg. Rank
1	A. Griezmann	CAM	2.33
2	L. Sané	LW	4.67
3	K. Mbappé	RM	5.00
4	Cristiano Ronaldo	ST	5.67
5	Coutinho	LW	5.67

Director’s note: Griezmann is the lead recommendation — top 3 on every metric. Sané and Mbappé are strong, high-confidence alternatives with the same pace-dribbling-finishing profile. Standardization was essential: it’s what separates a scouting tool from a transfer-value matcher.

5 Limitations & Data Notes

- Only 8 skills + Value used; Positioning, Vision, Crossing were out of scope.
- Age, contract status, injury history and league quality aren’t represented.
- Source data has minor string artifacts (e.g. “L. Modri?” literally appears in the raw CSV).
- 60 of 18,207 players dropped for missing Position/skill data, consistent with Task 2.2.